

One hacker... thousands of fake requests... and suddenly your entire backend crashes. How do companies like Netflix, Amazon, and Stripe protect their APIs from abuse?



What is Rate Limiting?



- Restricts how many requests a client can make
- Protects APIs from spam and abuse
- Prevents server overload



Why Rate Limiting is Important



- Protects against DDoS attacks
- Prevents brute-force login attempts
- Stops API abuse and bot traffic
- Improves system stability



What is a DDoS Attack?



- Massive number of fake requests sent to server
- Exhausts server resources
- Makes application slow or unavailable



How Rate Limiting Works?



- Server tracks incoming requests
- Request limit set per user/IP/API key
- Extra requests get blocked temporarily



Common Rate Limiting Rules



- 10 requests per minute
- 100 requests per hour
- 1000 requests per day



Different Rate Limiting Strategies



- IP-based limiting
- User-based limiting
- API key limiting
- Endpoint-specific limiting



Real-World Examples



- Login OTP protection
- Payment API protection
- Public API security
- AI API request limiting



Rate Limiting vs Throttling



Benefits of Rate Limiting



Common Mistakes Developers Make



Industry Best Practices

